

RepairLink

The Ultimate Master Operations & Knowledge Base

Comprehensive Documentation for Customers, Kaarigars, Admins, & Investors

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1. Executive Summary & Strategic Vision

RepairLink is an innovative, community-driven hyperlocal aggregator platform designed to bridge the severe gap between time-strapped urban consumers and skilled local repair artisans (Kaarigars). Conceived by Team Delta during the Spark Fellowship 2026, RepairLink serves as the foundational infrastructure for India's transition to a circular economy at the neighborhood level.

Our guiding philosophy is simple yet transformative: *"Mat Feko, Fix Karo"* (Don't throw it away, fix it). We aim to disrupt the deeply entrenched "buy-and-discard" culture by making "repair-and-reuse" not just an ideological choice, but the most frictionless, default, and economically viable option for urban residents.

To achieve this, we have engineered a comprehensive digital ecosystem that digitizes discovery, coordinates seamless doorstep logistics, enforces rigorous trust mechanisms, and introduces engaging gamification. This document acts as the definitive, all-encompassing knowledge base covering every facet of the RepairLink platform—from raw survey data to deep technical architecture, administrative protocols, and stakeholder FAQs. This corpus is specifically formatted and structured to empower RAG (Retrieval-Augmented Generation) AI models to resolve complex queries instantly and accurately.

2. The Problem Space & Market Catalyst

2.1 The "Buy-and-Discard" Loop

Modern urban consumption is trapped in a destructive cycle dictated entirely by convenience. When an everyday item—be it a pair of shoes, a mixer grinder, or a laptop—breaks down, consumers face multiple friction points that make buying a replacement significantly easier than

attempting a repair. Items sit in a "drawer economy" or are relegated to landfills simply because the cognitive and logistical load of finding a repairer outweighs the cost of replacement.

2.2 The Urban Demographic "Rahul" Persona

To ground our product design, we developed a data-backed user persona named "Rahul." Rahul is a 22-year-old early-career professional working at a startup, living in South/East Delhi. He represents our core demographic: digitally native, highly time-poor, sustainability-conscious, but ultimately constrained by the realities of urban logistics.

- **The Scenario:** Rahul's favorite, comfortable office shoes develop worn-out soles just days before an important business trip. He genuinely wants to keep them and fix them.
- **The Investigation:** He asks neighbors and colleagues for recommendations but receives vague directions ("There's a guy near the mother dairy market...").
- **The Decision Matrix:** He discovers the nearest reliable cobbler is a 15-minute walk away. With his heavy workload, carving out 45 minutes for a round trip to drop them off, plus another trip to pick them up later, is impossible.
- **The Result:** The friction is too high. Rahul abandons his repair intent, orders a new pair of shoes online for ₹2,500 via next-day delivery, and the old shoes are discarded. The buy-and-discard loop claims another victim.

2.3 The Three Core Frictions

Through our extensive quantitative survey (n=123 residents across Delhi-NCR), we pinpointed the exact structural failures of the offline repair market. These three core frictions form the absolute foundation of why RepairLink exists:

1. **The Discovery Gap (7.8%):** Only 7.8% of urban consumers actively know exactly where to find a reliable repair artisan. The vast majority (63.8%) suffer from a complete awareness deficit regarding local services within a 10-minute walk.
2. **The Convenience Barrier (75.9%):** 75.9% of users stated that doorstep pick-and-drop logistics is the single most critical factor that would persuade them to use a repair service. Time is the ultimate currency, and offline repair consumes too much of it.
3. **The Trust Deficit (42.2%):** Currently, 42.2% of users rely exclusively on family or neighbor word-of-mouth. Consumers are terrified of being scammed, overcharged, or receiving ruined items back. Without standardized reviews, verifiable before/after photos, or transparent pricing, consumers opt for the safety of buying new.

2.4 Market Timing & Policy Context

The window of opportunity to capture the hyperlocal repair market is narrow and opening rapidly, driven by three powerful catalysts:

1. **Right to Repair Movement (Est. 2022):** The Indian government's ongoing push for consumer protection and device repairability creates immense cultural tailwinds. While currently focused on large electronics, it legitimizes the broader repair economy and formalizes independent repair shops.
2. **The Urban Company Precedent:** Over the last decade, platforms like Urban Company and Blinkit have successfully conditioned Indian consumers to trust managed-local-services and pay convenience fees for doorstep delivery. However, the basic artisan repair category (cobblers, tailors, bag menders) remains completely unorganized and untouched.
3. **Corporate ESG Sustainability Pressures:** Stricter Extended Producer Responsibility (EPR) regulations and e-waste management rules are forcing brands to prioritize repairability. RepairLink is uniquely positioned to offer B2B partnerships for large-scale e-waste and textile waste diversion.

3. Exhaustive Market Research & Quantitative Data

Team Delta conducted a rigorous primary research survey across all 10 zones of Delhi-NCR to validate our hypotheses. The data below informs every UI, pricing, and operational decision made on the platform.

3.1 Demographics & Income Distribution

Our sample size was deeply targeted. The respondent base was predominantly young, with 81% falling within the 18–24 age bracket. Occupation-wise, 70.7% were students, while 13.8% were working professionals in the private sector.

Financially, the bulk of respondents belonged to small-to-mid households (55.2% living with 3-4 members). The defining income metric was a monthly household income oscillating between ₹50k and 1.5L. This defines the perfect sweet spot: consumers who possess enough discretionary income to afford a small convenience fee, yet are value-conscious enough to prefer a ₹200 repair over a 2,500 replacement.

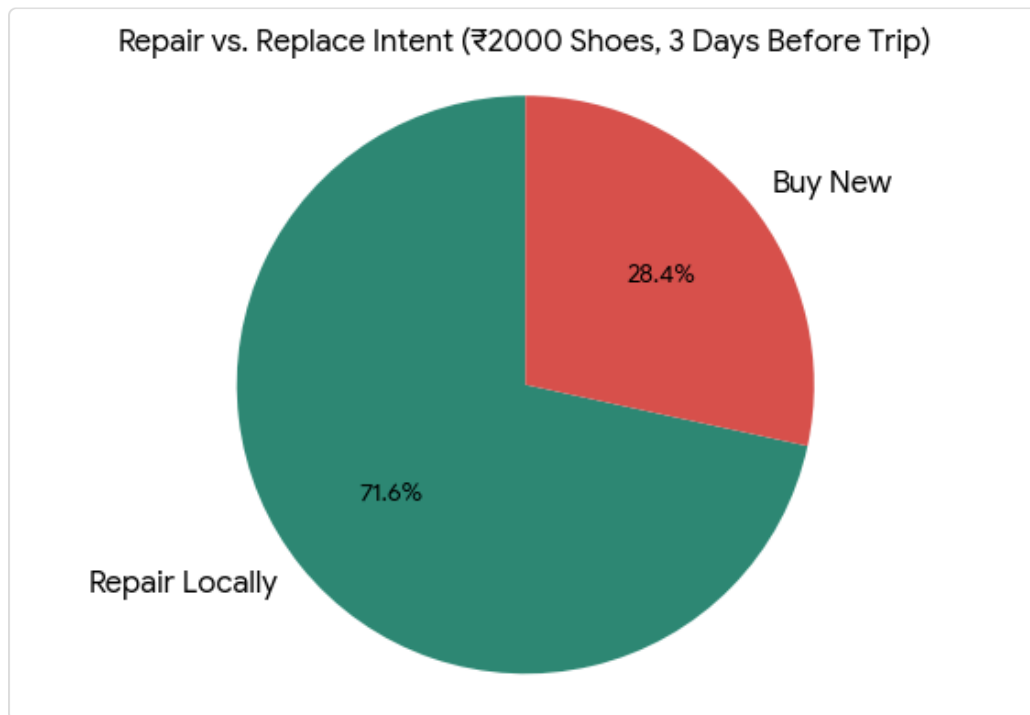
3.2 Discovery Gap Analysis

When asked how they locate local services, the data was stark:

- **Word of Mouth:** 62% rely purely on asking family or having a pre-existing contact. Active searching is virtually non-existent.
- **Hyperlocal Ignorance:** When asked if they knew the small shops within a 10-minute walk, only 37% claimed to know them "very well." Close to half admitted to relying heavily on convenience. If an option is more than 5-10 minutes away, 47% of users state that distance heavily influences them to abandon the task.

3.3 Repair vs. Replace Behavior & Intent

We tested pure repair intent using high-pressure psychological scenarios. We asked: *"Your ₹2,000 shoes just broke and you need them for a trip in 3 days. Would you repair them or buy new ones?"*

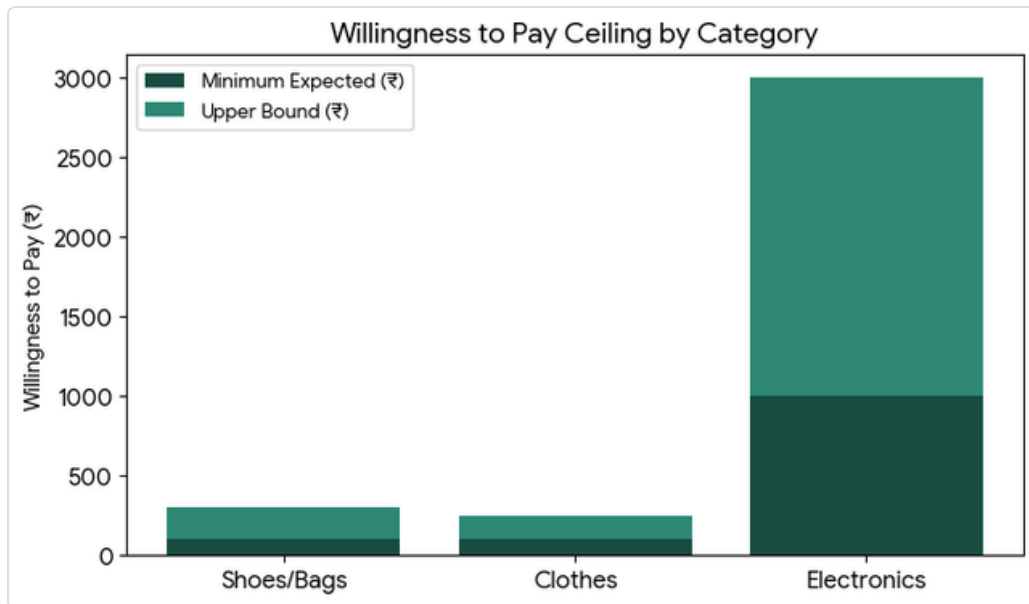


An overwhelming **71.6% chose to repair locally**, proving that the core intent to save money and salvage items is extremely resilient, even under time pressure. The bottleneck occurs solely because they do not know *who* to call.

Furthermore, when asked about the environmental factor, 39.7% of respondents viewed "reducing waste" as a nice bonus, while for 20.7%, it was the primary motivator. Sustainability is a strong secondary marketing lever, but convenience remains king.

3.4 Psychological Pricing & Willingness to Pay

We mapped the exact psychological price ceilings across three distinct repair categories to build our platform's tiered monetization strategy. We discovered that willingness to pay scales linearly with the avoidance cost (replacement value).



- **Shoes & Bags:** The absolute comfort zone lies between ₹100 and ₹300. Anything above this, and the user pivots to buying a new pair.
- **Clothing Alterations:** A highly casual category. Comfort zone sits tightly between ₹100 and ₹250.
- **Electronics (Laptops/Phones):** The ceiling shatters. Because replacing a laptop costs ₹40,000+, users are highly comfortable spending between ₹1,000 and ₹3,000 to salvage the device. Crucially, 36% of respondents cited electronics repair pricing as the "most unfair or confusing" offline experience today.

3.5 Zone-by-Zone Heatmap Data

Rather than guessing where to launch, we plotted behavioral data geographically. **South Delhi** and **East Delhi** registered the highest intersection of "convenience-driven behavior" and "willingness to pay for electronics repair." These two zones were subsequently selected for the Phase 1 MVP rollout.

4. Product & UX Strategy

RepairLink's UI and product architecture were explicitly designed to directly address the three frictions identified in the survey.

4.1 The 2-Tap Solution Workflow

1. **Digital Discovery (The Map Interface):** Upon opening the app, users are greeted with a map interface showing verified artisans within a 2-5 km radius. They can filter by trade (Cobbler, Locksmith, Tailor, Appliance Tech).
2. **Instant Quoting & Booking:** Users select the broken item category and pay a nominal, transparent platform connection fee.
3. **Doorstep Logistics:** Leveraging third-party APIs (or internal runners during the pilot), the item is picked up from the user's doorstep, delivered to the artisan, and returned within the SLA window.
4. **Digital Tracking:** The User Dashboard (`UserDashboard.html`) provides real-time updates: *Pending Payment → Kaarigar Assigned → Repair in Progress → Ready for Drop-off → Completed.*

4.2 The Trust Infrastructure (The Trust Layer)

To combat the 42.2% Trust Deficit, we built the "Trust Layer." This is a mandatory operational protocol enforced by the platform software:

- **Mandatory Before/After Photography:** The delivery runner takes a timestamped, geo-tagged photo of the broken item at pickup. The artisan must upload an "after" photo proving the fix before payment is released.
- **Live Turnaround Badges:** Artisans who consistently meet SLA deadlines earn automated "Fast Turnaround" badges on their public profile.
- **Transparent Reviews:** Upon order completion, an interactive star-rating modal is pushed to the user. This dynamic average feeds directly into the artisan's visibility algorithm.

5. Technical Architecture & Security

The RepairLink platform (v4.0 2026 Edition) relies on a highly optimized, lightweight technology stack designed for extreme speed and scalability.

5.1 Frontend & Database Stack

- **Frontend:** HTML5, CSS3, and Vanilla JavaScript. We utilize zero heavy frameworks (no React/Angular) to guarantee ultra-fast mobile loading speeds in variable network conditions.
- **UI Aesthetics:** The frontend utilizes advanced glassmorphism (frosted glass panels), smooth hover state animations, and dynamic pastel mesh gradients.
- **Backend & Database:** Google Firebase. We leverage Firebase Authentication for secure sign-ins and Cloud Firestore (NoSQL) for real-time order tracking and database management.
- **Hosting:** Deployed seamlessly via Firebase Hosting on the domain `repairlink-delta.web.app`.

5.2 Geo-Fenced Discovery & Location APIs

We use **Leaflet JS** integrated with **CARTO Voyager tiles** for our mapping interface. To ensure users do not book services outside our pilot operational zones, we developed an **Inverted Polygon Mask**. This programmatic mask brightly highlights the active service areas (e.g., North/South Delhi) while visually dimming and disabling clicks on unserviceable regional boundaries.

5.3 Security, XSS Mitigation, & Privacy

Data privacy is critical, particularly concerning the safety of local artisans.

- **Anti-Scraping Measures:** Kaarigar contact numbers and exact street coordinates are strictly masked from the public interface. They are unlocked server-side only after a verified user pays the upfront connection fee.
- **XSS Protection:** All user-generated inputs (such as repair item descriptions, addresses, and review text) are strictly escaped and sanitized before being injected into the DOM via innerHTML, preventing malicious script injections.

- **Strict RBAC (Role-Based Access Control):** Firebase security rules strictly segregate users, kaarigars, and admins. A standard user physically cannot read the entire list of artisan phone numbers or access admin analytics endpoints.

6. The Eco-Gamification Engine

A core part of our retention strategy is conditioning users to associate repair with a positive dopamine hit. To achieve this, we built a sophisticated Eco-Gamification Engine.

6.1 Waste Points Algorithm

For every completed repair, users are awarded **Waste Points**. The internal calculation assumes that each standard repair equates to approximately 1.2kg of waste diverted from a landfill and 3.5kg of CO2 emissions reduced. The User Dashboard dynamically visualizes these metrics using animated progress bars.

6.2 Progressive Levels & Badges

Users climb through ranks based on total Waste Points earned, which unlocks prestige levels visible on their profile:

Level Name	Points Required	User Status / Description
Eco-Beginner	0 - 499	Starting the circular economy journey.
Eco-Warrior	500 - 1499	Consistently repairing basic items.
Eco-Champion	1500 - 2999	Advocate for zero-waste lifestyle.
Eco-Master	3000+	Elite sustainability leader. (Unlocks Premium Perks)

In addition to levels, the UI grants interactive badges:

- 🌱 **Early Adopter:** Automatically granted to all accounts created during the 2026 Beta Phase.
- 🔧 **First Repair:** Unlocks instantly upon the status of the first order switching to 'completed'.
- 🏆 **Eco-Saver:** Unlocks automatically upon completing 5 distinct repair orders.

7. Platform Economics & Business Model

RepairLink monetizes software discovery and logistics utility, NOT the artisan's hard labor. We do not charge percentage commissions on the repair work itself.

7.1 Dynamic Connection Fees

When a user decides to hire an artisan, they must pay a highly transparent, non-refundable platform connection fee upfront. This fee covers our backend matching, UI utility, and the Trust Layer.

- **Cobbler / Shoe Repair:** ₹9 connection fee.
- **Tailor / Locksmith / Watch Repair:** ₹19 connection fee.
- **Electronics / Mechanics / Appliance Repair:** ₹39 connection fee.

Example: A user pays ₹9 to RepairLink via UPI on the app. The artisan arrives, quotes ₹150 for the shoe repair, does the job, and keeps 100% of the ₹150 directly from the customer.

7.2 Premium Subscription Tiers

To capture our "Convenience-First" demographic, we offer **RepairLink Premium**.

- **Price:** ₹59 for a 3-Month Plan (Discounted to ₹39 with promo code "DELTA").
- **Benefits:** Subscribers receive 3 "Free Service Requests" (the ₹9-₹39 connection fees are completely waived). They also receive priority matching algorithms and dedicated WhatsApp support.
- **Value Proposition:** The subscription pays for itself if the user books just 2 appliance repairs in 3 months.

7.3 Go-To-Market & Financial Projections

We are executing a highly disciplined, density-first 12-month rollout roadmap designed to reach 50,000 active users:

- **Phase 1 (Months 1-3): MVP & Validation.** Launch exclusively in South/East Delhi. Onboard 50 verified cobblers/bag menders. Target: 5,000 active users. Focus on perfecting the logistics margin (e.g., collecting the connection fee).
- **Phase 2 (Months 4-6): Scale & Series A.** Expand to 200 artisans. Launch the high-margin electronics repair category. Initiate B2B brand partnerships and close Series A funding (₹8-12 Cr target). Target: 20,000 users.
- **Phase 3 (Months 7-12): Consolidate & Expand.** City-wide rollout across Central & North Delhi (500+ artisans). Launch corporate e-waste portal. Approach operational break-even at 50,000+ users.

8. Customer Knowledge Base (B2C)

This section outlines workflows and policies for end-users accessing the platform.

8.1 Customer Dashboard Workflows

Upon authenticating via Firebase, users land on `UserDashboard.html`. From here, they can manage their entire repair lifecycle:

- **New Request Tab:** Users select a category (e.g., Appliance), describe the issue (e.g., "Mixer jar broken"), input their address, and scan the dynamic UPI QR code to pay the upfront connection fee.
- **Live Requests Tab:** Tracks order statuses (Pending Payment → Kaarigar Assigned → Repair in Progress → Completed). If an order sits in "Repair in Progress" for over 48 hours, an automated nudge button appears: *"Has your item been repaired? Mark as Completed."*
- **Rating Modal:** Once an order is marked done, an interactive 5-star modal forces the user to review the Kaarigar before dismissing the prompt, ensuring high data quality for the Trust Layer.

8.2 Customer FAQs

Q: What devices or items do you repair?

A: We repair all major electronics including smartphones, laptops, tablets, and smart home appliances (mixers, fans), in addition to our core artisan categories (shoes, bags, clothing alterations, and locksmith services).

Q: How long does a typical repair take?

A: It depends heavily on the category. Most smartphone screen and battery replacements take under 2 hours. Laptops and complex appliance repairs usually take 1-3 business days. Artisan repairs (shoes/clothes) are typically completed within 24-48 hours from pickup.

Q: Is there a warranty on repairs?

A: Yes, RepairLink enforces a 90-day warranty on all parts and labor for every repair officially tracked and completed through the RepairLink platform.

Q: What happens if an artisan inspects my item and says it cannot be fixed?

A: If an artisan determines an item is beyond repair, your connection fee is fully refunded to your wallet. Furthermore, we will assist you in safely recycling the item through our certified B2B e-waste partners.

Q: Can I negotiate the final repair price directly with the Kaarigar?

A: Absolutely. The repair estimates provided on our Pricing page are industry-average guidelines to set expectations. The final cost of labor and parts is negotiated and agreed upon directly between you and the Kaarigar. We only govern the initial platform connection fee.

Q: How do I earn and use Waste Points?

A: You earn Waste Points automatically every time a repair order is marked as 'Completed'. You can track your carbon footprint reduction in your dashboard. Earning 3000+ points grants you 'Eco-Master' status, which unlocks periodic free Premium perks.

Q: Is my payment secure?

A: Yes, all Premium connection fee payments are handled securely through verified, bank-grade UPI payment gateways and tracked immutably in your User Dashboard.

9. Kaarigar Knowledge Base (Partner Portal)

Our Kaarigars are the backbone of the platform. This section outlines the procedures for artisans interacting with RepairLink.

9.1 Onboarding & Verification Protocol

To maintain the integrity of the Trust Layer, we do not allow open registration for artisans. They must undergo a vetting process.

- **Application:** Artisans apply via the Partner Portal, submitting basic KYC documents and proof of trade.
- **Experience Requirement:** We require a minimum of 2 years of verifiable offline experience in their specific trade.
- **Assessment & Background Check:** Our local operations team conducts a standard physical assessment test and a background check. Parts must be sourced from verified OEMs where possible.
- **Approval:** Once verified, the Admin grants the user account 'Kaarigar' status via Firebase RBAC, granting them access to `KaarigarDashboard.html`

9.2 Kaarigar Dashboard & Earnings

The Kaarigar Dashboard serves as their digital storefront and CRM:

- **Profile Mapping:** Kaarigars must keep their location accurate. They can use the 1-click Auto-Detect GPS feature or drop a pin on an embedded Leaflet map to update their coordinates securely.
- **Order Management:** Incoming requests from users appear here. Kaarigars can accept requests, triggering the platform to reveal their contact data to the paying user.

- **Earnings & Reviews Suite:** A dedicated tab allows artisans to track their total completed jobs, read customer reviews to improve their service, and access direct Help & Support from the RepairLink ops team.

9.3 Kaarigar FAQs

Q: Does RepairLink take a commission or percentage cut from my repair earnings?

A: No. RepairLink operates on a zero-commission model for labor. You keep 100% of the money the customer pays you for the repair and parts. We make our money by charging the customer a small connection fee upfront.

Q: How do I get more repair jobs?

A: The platform's algorithm favors Kaarigars with high average star ratings and fast turnaround times. Consistently taking before/after photos and requesting your customers to leave a 5-star review on the app is the fastest way to rank higher.

Q: What if a customer refuses to pay me after I complete the repair?

A: Immediately raise a support ticket via the Help tab in your dashboard. The operations team will step in to mediate the dispute, and abusive customers will be permanently banned from the platform.

10. Administrator Knowledge Base

Administrators oversee the health, security, and financial integrity of the RepairLink ecosystem via Admin.html

10.1 Admin Dashboard Capabilities

The Admin Control Center provides a bird's-eye view of all platform metrics using dynamic, real-time charting components:

- **Live Revenue Tracking:** Automated calculation of connection fees collected and active premium subscriptions.
- **Traffic Analytics:** Tracks daily page visits across index, maps, pricing, and dashboards to analyze funnel drop-offs.
- **User & Order Management:** Monitors live counts of Standard vs. Premium users, active service distributions (e.g., identifying a surge in AC repairs during summer), and identifies stale orders.
- **Kaarigar Verification Pipeline:** Admins review uploaded KYC documents and manually click "Approve" or "Reject" to alter a user's RBAC status to 'kaarigar' in the Firestore database.

10.2 Conflict Resolution & Edge Cases

Admins handle critical edge cases:

- **Payment Verification (Status: pending_payment):** For users who pay via the static UPI QR code, admins must manually verify the transaction ID against the bank statement and flip the order status to "Kaarigar Assigned" (pending).
- **Abandoned Orders:** If an order remains "In Progress" for over 5 days with no action from the user or kaarigar, the admin will intervene, contact both parties via WhatsApp, and force-resolve or cancel the ticket to maintain database hygiene.
- **Trust Layer Violations:** If a Kaarigar fails to provide before/after photos on three consecutive orders, the admin system automatically flags their profile for manual review and possible suspension.

11. AI Integration (Llama-3 70b)

To ensure unparalleled scalability in customer support, RepairLink utilizes an advanced external AI Chatbot architecture.

- **Tech Stack:** We integrated a custom Llama-3 70b Large Language Model accessed via the ultra-fast Groq API. The interface is built using Python, Streamlit, and LangChain.
- **Deployment:** The chatbot exists as a floating, draggable widget seamlessly embedded directly into the platform interface.
- **Functionality:** The AI acts as a 24/7 autonomous support agent. It is strictly fine-tuned and trained on this exact knowledge base corpus. Its primary mandates are to enforce the "*Mat Feko, Fix Karo*" philosophy, assist users in navigating the UI, explain the intricacies of the Gamification Engine, and guide customers smoothly through the transparent connection fee payment process.

End of Document

RepairLink v4.0 - Internal Confidential Documentation